DESIGN SPECIFICATION

VSUART

[LOGO]

REVISION HISTORY			
DATE	**VERSION**	**DESCRIPTION**	**AUTHOR**
2026/03/03	0.01	Initial Version	

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1. Overview

This document describes the design of VSUART IP Core.

1.1 Functional overview

Refer to Functional Specification.

1.2 Difference from re-used modules

CDC (clock domain crossing) unit:

- Add handshake signal from higher to lower frequency domain

Asynchronous FIFO (TX FIFO, RX FIFO)

- Add signals displaying the number of remaining charaters in FIFO

1.3 Block diagram

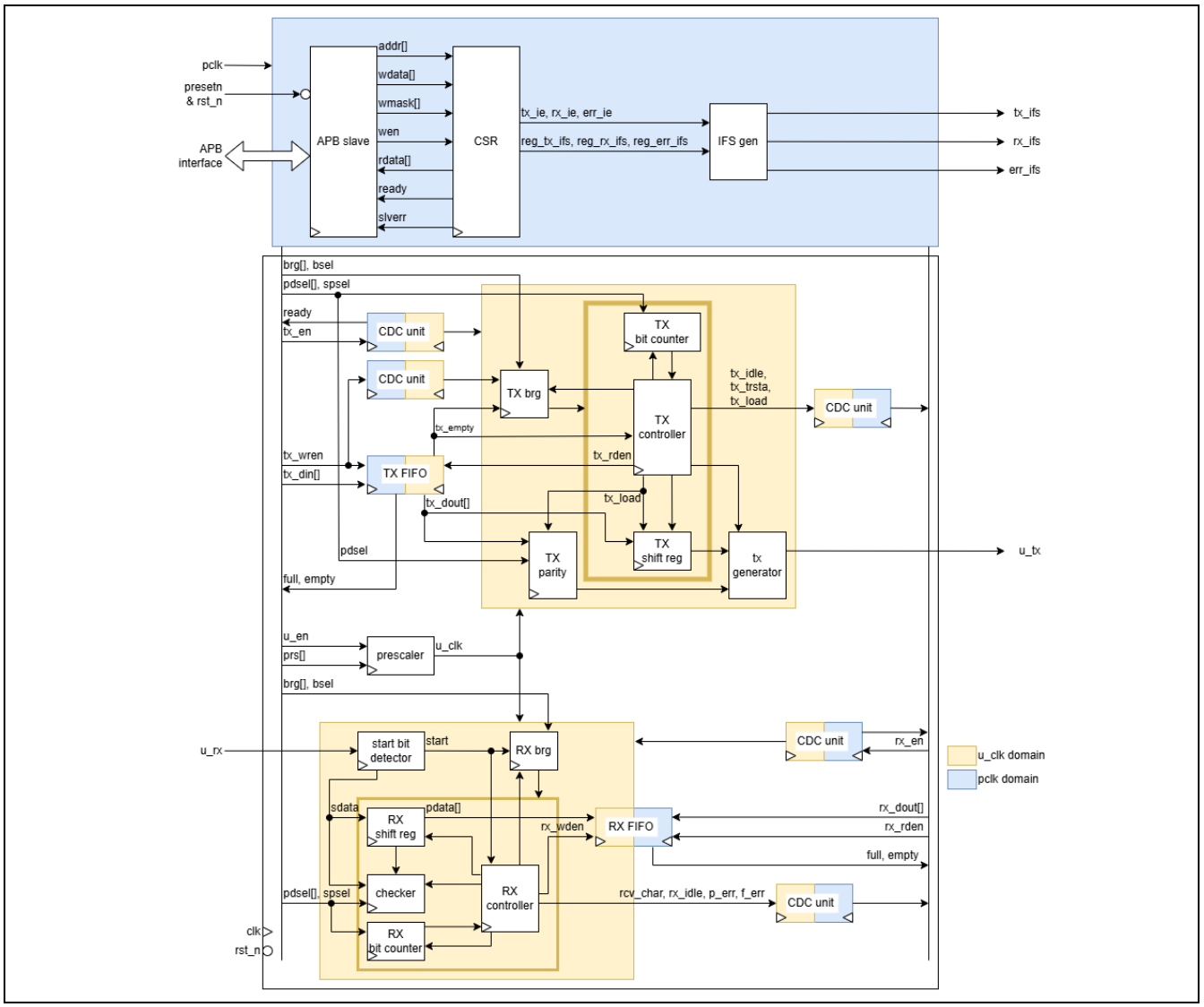


Figure 1-1: Block Diagram of VSUART

1.4 RTL Hierarchy

Block	Description
vsuart	
vsuart_prs	
vsuart_apb	
vsuart_apb_ctrl	
vsuart_apb_reg	
vsuart_apb_intr	
vsuart_tx	
vsuart_tx_brg	
vsuart_tx	
vsuart_tx_brg	

Table 1-1: RTL Hierarchy of VSUART

2. Signal description

The module input/output signals are defined in the Functional Specification.

The internal signals are listed below.

Name	Bus	From	To	Description	Init.
					-
					-

Table 2-1: Signal descriptions

3. Detailed description

3.1 vsuart_apb module: Level-1 Submodule

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3.1.1 Signal descriptions

Name	Bus	I/O	Init.	Active	Connection	Note
rst_n	-	I	-	0	TOP	-
clk	-	I	-	↑	TOP	-
i_pclk	-	I		↑	TOP	
i_presetn	-	I		0	TOP	
i_paddr	[15:0]	I		1	TOP	
i_psel	-	I		1	TOP	
i_penable	-	I		1	TOP	
i_pwrite	-	I		1	TOP	
i_pwdata	[31:0]	I		1	TOP	
i_pstrb	[3:0]	I		1	TOP	
		I				
		I				
o_pready		O	1	1	TOP	
o_prdata		O	0	1	TOP	
o_pslverr		O	0	1	TOP	

Table 2-1: I/O descriptions of vsuart_apb

Name	Bus	From	To	Description	Init.
					-
					-

Table 2-1: Internal signal descriptions of vsuart_apb

3.1.2 Block diagram

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3.1.3 Functional details

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3.2 vsuart_apb_ctrl module: Level-2 Submodule

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3.2.1 Signal descriptions

Name	Bus	I/O	Init.	Active	Connection	Note
rst_n	-	I	-	0	TOP	-
clk	-	I	-	↑	TOP	-

i_pclk	-	I		↑	TOP	
i_presetn	-	I		0	TOP	
i_paddr	[15:0]	I		1	TOP	
i_psel	-	I		1	TOP	
i_penable	-	I		1	TOP	
i_pwrite	-	I		1	TOP	
i_pwdata	[31:0]	I		1	TOP	
i_pstrb	[3:0]	I		1	TOP	
o_pready		O	1	1	TOP	
o_prdata		O	0	1	TOP	
o_pslverr		O	0	1	TOP	

Table 2-1: I/O descriptions of vsuart_apb

Name	Bus	From	To	Description	Init.
					-
					-

Table 2-1: Internal signal descriptions of vsuart_apb

3.2.2 Block diagram

3.2.3 Functional details